Chapter 1

Electronic Structure Calculations: Independent-Particle Models

Solving the time-independent Schrödinger equation:

$$\hat{H}_{tot}\Psi_{tot} = E_{tot}\Psi_{tot}. \tag{1.1}$$

where

$$\begin{aligned}\hat{H}_{tot} &= -\sum_{A=1}^{M}\frac{\hbar^2}{2M_A}\nabla_A^2 - \sum_{i=1}^{N}\frac{\hbar^2}{2m_e}\nabla_i^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{1}{4\pi\epsilon_0}\frac{Z_A e^2}{r_{iA}} \\ &\quad + \sum_{i=1}^{N}\sum_{j>i}^{N}\frac{1}{4\pi\epsilon_0}\frac{e^2}{r_{ij}} + \sum_{A=1}^{M}\sum_{B>A}^{M}\frac{1}{4\pi\epsilon_0}\frac{Z_A Z_B}{R_{AB}}.\end{aligned} \tag{1.2}$$

1.1 Independent Particle Models

1. In independent particle models, the motion of one electron is considered to be independent of the dynamics of all other electrons.

2. Two ways of approximating the interactions between the particles

(a) by neglecting all but the most important one. → not acceptable in accuracy within electronic structure theory.

(b) by taking all interactions into account in an average fashion. → acceptable in accuracy within electronic structure theory. → Hartree-Fock (HF theory)

1.2 Outlines of the HF theory

1. One electron is described by an **orbital**.
2. The total wave function is given as **a product of orbitals**.
3. Since electrons are indistinguishable fermions (particles with a spin of 1/2), the total wave function must be **antisymmetric → Slater determinant**.
4. The best set of orbitals is determined by the **variational principle**.
5. Since the interaction of one electron to all the other electrons is included in an average way and the other electrons are described by their respective orbitals, the HF equations depend on their own solutions, and must therefore be solved iteratively (**Self-Consistent Field (SCF) method**).

$$\hat{H}_{tot}\Psi_{tot} = E_{tot}\Psi_{tot}$$

$$\downarrow \qquad \Psi_{tot} = \textit{single determinant}$$

HF equations

Additional approximation	↙↘	*Addition of* more determinants
Semi-empirical methods		Convergence to exact solution

(1.3)

1.3 Post-HF Theories

1. HF theory only accounts for the average electron-electron interactions, and consequently neglects **the correlation between electrons**.

2. Methods that include electron correlation require a **multi-determinant wave functions**, since HF is the best single-determinant wave function. → Configuration Interaction (CI), Perturbation Theory (PT), and Coupled Cluster (CC) theory.

   (a) dynamic correlation: important in van der Waals interaction.

   (b) static (non-dynamic) correlation: included in multi-configurations (or references).

3. **Density Functional Theory (DFT)** in the Kohn-Sham version can be considered as an improvement on HF theory, where the many-body effect of electron correlation is modeled by a function of the electron density.

   (a) DFT is an independent-particle model, and is comparable to HF computationally, but provides significantly better results.

   (b) The main disadvantage of DFT is that there is no systematic approach to improving the results towards the exact solution.

1.4 The Electronic Problem

1.4.1 Non-Relativistic Time-Independent Schrödinger Equation

In the non-relativistic time-independent Schrödinger equation,

$$\hat{H}_{tot}\Psi_{tot} = E_{tot}\Psi_{tot}. \tag{1.4}$$

the total Hamiltonian operator can be written as kinetic and potential energies of the nuclei and electrons:

$$\begin{aligned}\hat{H}_{tot} &= -\sum_{A=1}^{M}\frac{\hbar^2}{2M_A}\nabla_A^2 - \sum_{i=1}^{N}\frac{\hbar^2}{2m_e}\nabla_i^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{1}{4\pi\epsilon_0}\frac{Z_A e^2}{r_{iA}} \\ &\quad + \sum_{i=1}^{N}\sum_{j>i}^{N}\frac{1}{4\pi\epsilon_0}\frac{e^2}{r_{ij}} + \sum_{A=1}^{M}\sum_{B>A}^{M}\frac{1}{4\pi\epsilon_0}\frac{Z_A Z_B}{R_{AB}}. \end{aligned} \tag{1.5}$$

1.4.2 Atomic Units

To cast Eq. (1.5) into dimensionless form, we let $X, Y, Z \rightarrow \lambda X', \lambda Y', \lambda Z'$ for nuclear coordinates and $x, y, z \rightarrow \lambda x', \lambda y', \lambda z'$ and obtain

$$\begin{aligned}\hat{H}_{tot} &= -\sum_{A=1}^{M}\frac{\hbar^2}{2M_A\lambda^2}\nabla_A^2 - \sum_{i=1}^{N}\frac{\hbar^2}{2m_e\lambda^2}\nabla_i^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{1}{4\pi\epsilon_0}\frac{Z_A e^2}{\lambda r_{iA}} \\ &\quad + \sum_{i=1}^{N}\sum_{j>i}^{N}\frac{1}{4\pi\epsilon_0}\frac{e^2}{\lambda r_{ij}} + \sum_{A=1}^{M}\sum_{B>A}^{M}\frac{1}{4\pi\epsilon_0}\frac{Z_A Z_B}{\lambda R_{AB}}. \end{aligned} \tag{1.6}$$

We choose λ such that

$$\frac{\hbar^2}{m_e\lambda^2} = \frac{e^2}{4\pi\epsilon_0\lambda} \equiv E_a, \tag{1.7}$$

where E_a is the atomic unit of energy called the *Hartree*. Solving Eq. (1.7) for λ, we find

$$\lambda = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} = a_0. \tag{1.8}$$

λ is just the Bohr radius a_0 which is the atomic unit of length called a *Bohr*. Finally, since Eq. (1.6) can be written as

$$\begin{aligned}\hat{H}_{tot} &= E_a\Bigg[-\sum_{A=1}^{M}\frac{1}{2(M_A/m_e)}\nabla_A'^2 - \sum_{i=1}^{N}\frac{1}{2}\nabla_i'^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{Z_A}{r'_{iA}} \\ &\quad + \sum_{i=1}^{N}\sum_{j>i}^{N}\frac{1}{r'_{ij}} + \sum_{A=1}^{M}\sum_{B>A}^{M}\frac{Z_A Z_B}{R'_{AB}}\Bigg], \end{aligned} \tag{1.9}$$

if we let $\hat{H}'_{tot} = \hat{H}_{tot}/E_a$, we obtain the dimensionless Schrödinger equation

$$\begin{aligned} \hat{H}'_{tot} &= -\sum_{A=1}^{M} \frac{1}{2M'_A} \nabla'^2_A - \sum_{i=1}^{N} \frac{1}{2} \nabla'^2_i - \sum_{i=1}^{N} \sum_{A=1}^{M} \frac{Z_A}{r'_{iA}} \\ &\quad + \sum_{i=1}^{N} \sum_{j>i}^{N} \frac{1}{r'_{ij}} + \sum_{A=1}^{M} \sum_{B>A}^{M} \frac{Z_A Z_B}{R'_{AB}}. \end{aligned} \tag{1.10}$$

Chapter 2

Hartree-Fock (HF) Approximation: Self-Consistent Field method

2.1 Variational Method

2.1.1 Variational Principle

The variational principle states that if *psi* is any well-behaved function that satisfies the boundary conditions associated with the problem of interest, then the expectation value of $\hat{H}$, calculated using ψ, will obey the inequality

$$E_{trial} = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} \geq E_0 \tag{2.1}$$

where E_0 is the exact ground state energy. The *psi* is called trial wavefunction.

To prove the theorem, we rearrange Eq. (2.1) to yield the inequality:

$$I = \langle\psi|\hat{H} - E_0|\psi\rangle \geq 0. \tag{2.2}$$

We expand *psi* in terms of the exact eigenfunctions ϕ of $\hat{H}$ which gives

$$\psi = \sum_k c_k \phi. \tag{2.3}$$

If Eq. (2.2) is substituted into Eq. (2.2), we get

$$I = \sum_k \sum_l c_k^* c_l \langle \phi_k | \hat{H} - E_0 | \phi_l \rangle = \sum_k \sum_l c_k^* c_l (E_k - E_0) \delta_{kl} = \sum_k |c_k|^2 (E_k - E_0) \tag{2.4}$$

Since $|c_k|^2 \geq 0$ and $E_k \geq E_0$, $I > 0$.

2.1.2 Linear Variational Method (The Rayleigh-Ritz Method)

1. Linear combination of atomic orbical-molecular orbital (LCAO-MO)): A molecular orbial can be expressed as a linear combination of n linearly independent atomic orbitals.

$$\psi = c_1\phi_1 + c_2\phi_2 + \cdots + c_n\phi_n = \sum_{j=1}^{n} c_j \phi_j \tag{2.5}$$

The expectation value $\langle E \rangle$ by ψ is then given by

$$\langle E \rangle = W = \frac{\int \psi^* \hat{H} \psi d\tau}{\int \psi^* \psi d\tau} = \frac{\sum_{j=1}^n \sum_{k=1}^n c_j c_k \int \phi_j^* \hat{H} \phi_k d\tau}{\sum_{j=1}^n \sum_{k=1}^n c_j c_k \int \phi_j^* \phi_k d\tau} = \frac{\sum_{j=1}^n \sum_{k=1}^n c_j c_k H_{jk}}{\sum_{j=1}^n \sum_{k=1}^n c_j c_k S_{jk}}, \tag{2.6}$$

where

$$H_{jk} = \int \psi^* \hat{H} \psi d\tau \tag{2.7}$$

$$S_{jk} = \int \psi^* \psi d\tau. \tag{2.8}$$

Eq. (2.6) can be written by

$$W \sum_{j=1}^{n} \sum_{k=1}^{n} c_j c_k S_{jk} = \sum_{j=1}^{n} \sum_{k=1}^{n} c_j c_k H_{jk} \tag{2.9}$$

W is a function of the n independent varialbes $c_1, c_2, \cdots, c_n$:

$$W = W(c_1, c_2, \cdots c_n) \tag{2.10}$$

A necessary condition for a minimum in a function W of several variables is that its partial derivatives with respect to each of the variables must be zero at the minimum point:

$$\frac{\partial W}{\partial c_i} = 0 \qquad (i = 1, 2, \cdots, n) \tag{2.11}$$

Differenciating Eq. (2.6) with respect to c_i to obtain n equations

$$\begin{aligned} & \frac{\partial W}{\partial c_i} \sum_{j=1}^{n}\sum_{k=1}^{n} c_j c_k S_{jk} + W \frac{\partial}{\partial c_i} \sum_{j=1}^{n}\sum_{k=1}^{n} c_j c_k S_{jk} = \frac{\partial}{\partial c_i} \sum_{j=1}^{n}\sum_{k=1}^{n} c_j c_k H_{jk} \\ \Rightarrow \quad & 2W \sum_{k=1}^{n} c_k S_{ik} = 2 \sum_{k=1}^{n} c_k H_{ik} \\ \Rightarrow \quad & \sum_{k=1}^{n} \left[H_{ik} - S_{ik} W c_k \right] = 0 \end{aligned} \tag{2.12}$$

Eq. (2.12) can be expressed as

$$\begin{pmatrix} H_{11} - S_{11}W & H_{12} - S_{12}W & H_{13} - S_{13}W & \cdots & H_{1n} - S_{1n}W \\ H_{12} - S_{12}W & H_{22} - S_{22}W & H_{23} - S_{23}W & \cdots & H_{2n} - S_{2n}W \\ \vdots & & & & \vdots \\ H_{n1} - S_{1n}W & H_{n2} - S_{n2}W & H_{n3} - S_{n3}W & \cdots & H_{nn} - S_{nn}W \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = 0 \tag{2.13}$$

or,

$$\begin{pmatrix} H_{11} & H_{12} & H_{13} & \cdots & H_{1n} \\ H_{12} & H_{22} & H_{23} & \cdots & H_{2n} \\ \vdots & & & & \vdots \\ H_{n1} & H_{n2} & H_{n3}W & \cdots & H_{nn} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} S_{11}W & S_{12}W & S_{13}W & \cdots & S_{1n}W \\ S_{12}W & S_{22}W & S_{23}W & \cdots & S_{2n}W \\ \vdots & & & & \vdots \\ S_{1n}W & S_{n2}W & S_{n3}W & \cdots & S_{nn}W \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

2. Eigenvalues and eigenvectors:

$$\mathbf{Hc} = E\mathbf{c}, \tag{2.14}$$

E are eigenvalues of $\mathbf{H}$ and $\mathbf{c}$ are eigenvectors of $\mathbf{H}$. (2.14) can be written by

$$\mathbf{Hc} - E\mathbf{c} = \mathbf{Hc} - E\mathbf{Ic} = (\mathbf{H} - E\mathbf{I})\mathbf{c} = \mathbf{0}, \tag{2.15}$$

or,

$$\begin{pmatrix} H_{11} - E & H_{12} & H_{13} & \cdots & H_{1n} \\ H_{12} & H_{22} - E & H_{23} & \cdots & H_{2n} \\ \vdots & & & & \vdots \\ H_{1n} & H_{n2} & H_{n3} & \cdots & H_{nn} - E \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = 0 \tag{2.16}$$

3. secular determinant and secular equation For there to be a solution to the Eq. (2.16) besides the trivial solution $0 = c_1 = c_2 = \cdots = c_n$, the determinant of $\mathbf{H} - E\mathbf{I}$ must vanish:

$$\begin{vmatrix} H_{11} - E & H_{12} & H_{13} & \cdots & H_{1n} \\ H_{12} & H_{22} - E & H_{23} & \cdots & H_{2n} \\ \vdots & & & & \vdots \\ H_{1n} & H_{n2} & H_{n3} & \cdots & H_{nn} - E \end{vmatrix} = 0 \tag{2.17}$$

The determinant of $\mathbf{H} - E\mathbf{I}$ is called **secular determinant** and (2.17) is called **secular equation**. (2.17) can be written by:

$$E^n + a_1 E^{n-1} + a_2 E^{n-2} + \cdots + a_{n-1} E + a_n = 0. \tag{2.18}$$

2.2 Electronic Wave Functions

2.2.1 The Antisymmetry or Pauli Exclusion Principle

1. electron spins: $|\alpha\rangle$: spin up, $|\beta\rangle$: spin down.

$$\int \mathrm{d}\omega \alpha^*(\omega)\alpha(\omega) = \int \mathrm{d}\omega \beta^*(\omega)\beta(\omega) = 1, \tag{2.19a}$$

$$\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1, \tag{2.19b}$$

$$\int \mathrm{d}\omega \alpha^*(\omega)\beta(\omega) = \int \mathrm{d}\omega \beta^*(\omega)\alpha(\omega) = 0, \tag{2.19c}$$

$$\langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle = 0. \tag{2.19d}$$

2. $\mathbf{x}$: spatial coordinates $\mathbf{r}$ plus spin coordinate ω of an electron.

3. $\Phi(\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N)$: The wave function for an N-electron system.

4. Antisymmetry or Pauli exclusion principle: A many-electron wave function must be antisymmetric with respect to the interchange of the coordinate $\mathbf{x}$ (both space and spin) of any two electrons,

$$\Phi(\mathbf{x}_1, \cdots, \mathbf{x}_i, \cdots, \mathbf{x}_j, \cdots, \mathbf{x}_N) = -\Phi(\mathbf{x}_1, \cdots, \mathbf{x}_j, \cdots, \mathbf{x}_i, \cdots, \mathbf{x}_N) \tag{2.20}$$

2.2.2 Spin-Orbitals and Spatial Orbitals

1. orbital: a wave function for a single electron.

2. molecular orbital (MO): a wave function for a single electrons in a molecule.

3. spatial orbital $\phi_i(\mathbf{r})$: a function of the position vector that describes the spatial distribution of an electron.

   (a) $\int \mathrm{d}\mathbf{r} \psi_i^*(\mathbf{r})\psi_j(\mathbf{r}) = \delta_{ij}$.

(b) complete set: $f(\mathbf{r}) = \sum_{i=1}^{\infty} a_i \psi_i(\mathbf{r})$.

4. spin orbital $\chi(\mathbf{x})$: the wave function for an electron that describes both its spatial distribution and its spin.

   (a) $\int \mathrm{d}\mathbf{x} \chi_i^*(\mathbf{r}) \chi_j(\mathbf{r}) = \delta_{ij}$.

2.2.3 Hartree Products

If we neglect electron-electron repulsion or consider an effective one-electron potential, the full electronic Hamiltonian becomes

$$\hat{H}_{el} = \sum_{i=1}^{N} \hat{h}(i). \tag{2.21}$$

When the operator $\hat{h}(i)$ has a set of spin orbitals as a set of eigenfunctions, that is,

$$\hat{h}(i)\chi_j(\mathbf{x}_i) = \epsilon_j \chi_j(\mathbf{x}_i), \tag{2.22}$$

a simple product of spin orbital wave functions for each electron, called a **Hartree product**,

$$\Psi^{HP}(\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N) = \chi_i(\mathbf{x}_1)\chi_2(\mathbf{x}_2)\cdots\chi_k(\mathbf{x}_N), \tag{2.23}$$

is an eigenfunction for the Schrödinger equation,

$$\hat{H}_{el}\Psi^{HP} = E\Psi^{HP}. \tag{2.24}$$

with eigenvalue E, which is just sum of the spin orbital energies,

$$E = \epsilon_i + \epsilon_j + \cdots + \epsilon_k. \tag{2.25}$$

The Hartree product does not satisfy the antisymmetry principle.

2.3 Hartree-Fock Approximation

2.3.1 Slater Determinants

Slater determinants is antisymmetric wave functions for N electrons occupying N spin orbitals $(\chi_i, \chi_j, \cdots, \chi_k)$ without specifying with electron is in which orbital:

$$\begin{aligned}\Psi(\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N) &= (N!)^{-1/2} \begin{pmatrix} \chi_i(\mathbf{x}_1) & \chi_j(\mathbf{x}_1) & \cdots & \chi_k(\mathbf{x}_1) \\ \chi_i(\mathbf{x}_2) & \chi_j(\mathbf{x}_2) & \cdots & \chi_k(\mathbf{x}_2) \\ \vdots & \vdots & & \vdots \\ \chi_i(\mathbf{x}_N) & \chi_j(\mathbf{x}_N) & \cdots & \chi_k(\mathbf{x}_N) \end{pmatrix}, \quad (2.26) \\ &= |\chi_i(\mathbf{x}_1)\chi_j(\mathbf{x}_2)\cdots\chi_i(\mathbf{x}_N)\rangle. \quad (2.27)\end{aligned}$$

where $(N!)^{-1/2}$ is a normalization constant and only the diagonal elements are written in Eq. (2.27).

2.3.2 Antisymmetrizing operator

In the Hartree-Fock approximation, the Slater determinant is give as

$$\begin{aligned}\Psi(\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N) &= \hat{A}\Psi^{HP}(\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N) \\ &= \hat{A}\chi_1(\mathbf{x}_1)\chi_2(\mathbf{x}_2)\cdots\chi_N(\mathbf{x}_N), \quad (2.28)\end{aligned}$$

where an antisymmetrizing operator $\hat{A}$ is defined as

$$\hat{A} = \frac{1}{\sqrt{N!}} \sum_{p=0}^{N-1} (-1)^p \hat{P} = \frac{1}{\sqrt{N!}} [\hat{1} - \sum_{ij} \hat{P}_{ij} + \sum_{ijk} \hat{P}_{ijk} - \cdots]. \quad (2.29)$$

The $\hat{1}$ is the identity operator, while the sum over $\hat{P}_{ij}$ generates all possible permutations of two electron coordinates, the sum over $\hat{P}_{ijk}$ all possible permutations of three electron coordinates, etc. The operator $\hat{A}$ commutes with $\hat{H}_{el}$, and $\hat{A}$ acting twice gives the same

as $\hat{A}$ acting once, multiplied with $\sqrt{N!}$:

$$\hat{A}\hat{H}_{el} = \hat{H}_{el}\hat{A}$$
$$\hat{A}\hat{A} = \sqrt{N!}\hat{A}. \tag{2.30}$$

EXAMPLE 1.

$$\begin{aligned}
\hat{A}\chi_1(\mathbf{x}_1)\chi_2(\mathbf{x}_2)\chi_3(\mathbf{x}_3) &= \hat{A}\chi_1(1)\chi_2(2)\chi_3(3), \\
&= \frac{1}{\sqrt{3!}}\big[\hat{1} - \sum_{ij}\hat{P}_{ij} + \sum_{ijk}\hat{P}_{ijk}\big]\chi_1(1)\chi_2(2)\chi_3(3), \\
&= \frac{1}{\sqrt{6}}\big[\chi_1(1)\chi_2(2)\chi_3(3) - \chi_1(2)\chi_2(1)\chi_3(3) - \chi_1(1)\chi_2(3)\chi_3(2) \\
&\quad -\chi_1(3)\chi_2(2)\chi_3(1) + \chi_1(3)\chi_2(1)\chi_3(2) + \chi_1(2)\chi_2(3)\chi_3(1)\big],
\end{aligned}$$

EXAMPLE 2.

$$\begin{aligned}
\hat{A}\hat{A}\chi_1(1)\chi_2(2)\chi_3(3) &= \hat{A}\Big\{\frac{1}{\sqrt{6}}\big[\chi_1(1)\chi_2(2)\chi_3(3) - \chi_1(2)\chi_2(1)\chi_3(3) - \chi_1(1)\chi_2(3)\chi_3(2) \\
&\quad -\chi_1(3)\chi_2(2)\chi_3(1) + \chi_1(3)\chi_2(1)\chi_3(2) + \chi_1(2)\chi_2(3)\chi_3(1)\big]\Big\}, \\
&= \chi_1(1)\chi_2(2)\chi_3(3) - \chi_1(2)\chi_2(1)\chi_3(3) - \chi_1(1)\chi_2(3)\chi_3(2) \\
&\quad -\chi_1(3)\chi_2(2)\chi_3(1) + \chi_1(3)\chi_2(1)\chi_3(2) + \chi_1(2)\chi_2(3)\chi_3(1), \\
&= \sqrt{6}\hat{A}\chi_1(1)\chi_2(2)\chi_3(3)
\end{aligned} \tag{2.31}$$

2.3.3 The Electronic Energy Calculation and Coulomb and Exchange Integrals

The total Hamiltonian operator can be written as

$$\hat{H}_{tot} = \hat{T}_{nu} + \hat{H}_{el}, \tag{2.32}$$

where

$$\hat{H}_{el} = \hat{T}_{el} + \hat{V}_{ne} + \hat{V}_{ee} + \hat{V}_{nn}. \tag{2.33}$$

Here

$$\begin{aligned}
\hat{T}_{el} &= -\sum_{i=1}^{N} \frac{1}{2}\nabla_i^2 \\
\hat{V}_{ne} &= -\sum_{i=1}^{N}\sum_{A=1}^{M} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} \\
\hat{V}_{ee} &= \sum_{i=1}^{N}\sum_{j>i}^{N} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \\
\hat{V}_{nn} &= \sum_{A=1}^{M}\sum_{B>A}^{M} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|}.
\end{aligned} \tag{2.34}$$

Now we define the one- and two- electron operators as

$$\begin{aligned}
\hat{h}_i &= -\frac{1}{2}\nabla_i^2 - \sum_{A=1}^{M} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} \\
\hat{g}_{ij} &= \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|},
\end{aligned} \tag{2.35}$$

where the one-electron operator $\hat{h}_i$ describes the motion of electron i in the field of all the nuclei, and the two electron operator $\hat{g}_{ij}$ describes the electron-electron repulsion. Then, the electronic Hamiltonian $\hat{H}_{el}$ is written as

$$\hat{H}_{el} = \sum_{i=1}^{N} \hat{h}_i + \sum_{i=1}^{N}\sum_{j>i}^{N} \hat{g}_{ij} + \hat{V}_{nn}. \tag{2.36}$$

The electronic energy is now written as

$$\begin{aligned}
E_{el} &= \langle\Psi|\hat{H}_{el}|\Psi\rangle, \\
&= \langle\hat{A}\Psi^{HP}|\hat{H}_{el}|\hat{A}\Psi^{HP}\rangle, \\
&= \sqrt{N!}\langle\Psi^{HP}|\hat{H}_{el}|\hat{A}\Psi^{HP}\rangle, \\
&= \sum_{p=0}^{N-1}\langle\Psi^{HP}|\hat{H}_{el}|\hat{P}\Psi^{HP}\rangle, \\
&= \sum_{p=0}^{N-1}\Bigg[\sum_{i=1}^{N}\langle\Psi^{HP}|\hat{h}_i|\hat{P}\Psi^{HP}\rangle + \sum_{i=1}^{N}\sum_{j>i}^{N}\langle\Psi^{HP}|\hat{g}_{ij}|\hat{P}\Psi^{HP}\rangle \\
&\quad + \langle\Psi^{HP}|V_{nn}|\hat{P}\Psi^{HP}\rangle\Bigg].
\end{aligned} \tag{2.37}$$

The nuclear repulsion operator in the last term of Eq. (2.37) is independent of electronic coordinates and can immediately be integrated to yield a constant

$$\langle \Psi^{HP} | V_{nn} | \hat{P} \Psi^{HP} \rangle = V_{nn} \langle \Psi^{HP} | \hat{P} \Psi^{HP} \rangle = V_{nn}. \tag{2.38}$$

For the one-electron operator, only the identity operator can give a non-zero contribution. For the coordinate of electron 1, this yields

$$\begin{aligned}
\langle \Psi^{HP} | \hat{h}_1 | \hat{P} \Psi^{HP} \rangle &= \langle \chi_1(1)\chi_2(2)\cdots\chi_N(N) | \hat{h}_1 | \hat{P}\chi_1(1)\chi_2(2)\cdots\chi_N(N) \rangle, \\
&= \langle \chi_1(1)\chi_2(2)\cdots\chi_N(N) | \hat{h}_1 | \chi_1(1)\chi_2(2)\cdots\chi_N(N) \rangle \\
&\quad - \langle \chi_1(1)\chi_2(2)\cdots\chi_N(N) | \hat{h}_1 | \chi_1(2)\chi_2(1)\cdots\chi_N(N) \rangle + \cdots, \\
&= \langle \chi_1(1) | \hat{h}_1 | \chi_1(1) \rangle \langle \chi_2(2) | \chi_2(2) \rangle \cdots \langle \chi_N(N) | \chi_N(N) \rangle \\
&\quad - \langle \chi_1(1) | \hat{h}_1 | \chi_2(1) \rangle \langle \chi_2(2) | \chi_1(2) \rangle \cdots \langle \chi_N(N) | \chi_N(N) \rangle + \cdots \\
&= \langle \chi_1(1) | \hat{h}_1 | \chi_1(1) \rangle = h_1^{'}.
\end{aligned} \tag{2.39}$$

For the two-electron operator in the second term of Eq. (2.37), only the identity and $\hat{P}_{ij}$ operators can give non-zero contributions. A three-electron permutation will again give a least one overlap integral between two different MOs, which will be zero. The term

arising from the identity operator is given by

$$\begin{aligned}
\langle \Psi^{HP}|\hat{g}_{12}|\hat{P}\Psi^{HP}\rangle &= \langle \chi_1(1)\chi_2(2)\chi_3(3)\cdots\chi_N(N)|\hat{g}_{12}|\hat{P}\chi_1(1)\chi_2(2)\chi_3(3)\cdots\chi_N(N)\rangle, \\
&= \langle \chi_1(1)\chi_2(2)\chi_3(3)\cdots\chi_N(N)|\hat{g}_{12}|\chi_1(1)\chi_2(2)\chi_3(3)\cdots\chi_N(N)\rangle \\
&\quad -\langle \chi_1(1)\chi_2(2)\chi_3(3)\cdots\chi_N(N)|\hat{g}_{12}|\chi_1(2)\chi_2(1)\chi_3(3)\cdots\chi_N(N)\rangle \\
&\quad -\langle \chi_1(1)\chi_2(2)\chi_3(3)\cdots\chi_N(N)|\hat{g}_{12}|\chi_1(3)\chi_2(2)\chi_3(1)\cdots\chi_N(N)\rangle + \cdots, \\
&= \langle \chi_1(1)\chi_2(2)|\hat{g}_{12}|\chi_1(1)\chi_2(2)\rangle\langle\chi_3(3)|\chi_3(3)\rangle\cdots\langle\chi_N(N)|\chi_N(N)\rangle \\
&\quad -\langle \chi_1(1)\chi_2(2)|\hat{g}_{12}|\chi_2(1)\chi_1(2)\rangle\langle\chi_3(3)|\chi_3(3)\rangle\cdots\langle\chi_N(N)|\chi_N(N)\rangle \\
&\quad -\langle \chi_1(1)\chi_2(2)|\hat{g}_{12}|\chi_3(1)\chi_2(2)\rangle\langle\chi_3(3)|\chi_1(3)\rangle\cdots\langle\chi_N(N)|\chi_N(N)\rangle + \cdots \\
&= \langle \chi_1(1)\chi_2(2)|\hat{g}_{12}|\chi_1(1)\chi_2(2)\rangle - \langle \chi_1(1)\chi_2(2)|\hat{g}_{12}|\chi_2(1)\chi_1(2)\rangle \\
&= J'_{12} - K'_{12}, \qquad (2.40)
\end{aligned}$$

where a Coulomb and exchange integrals J'_{12} and K'_{12} are defined as

$$J'_{12} = \langle \chi_1(1)\chi_2(2)|\hat{g}_{12}|\chi_1(1)\chi_2(2)\rangle, \qquad (2.41)$$

$$K'_{12} = \langle \chi_1(1)\chi_2(2)|\hat{g}_{12}|\chi_2(1)\chi_1(2)\rangle. \qquad (2.42)$$

The Coulomb integral represents the classical repulsion between two charge distributions described by $\chi_1^2(1)$ and $\chi_2^2(2)$. The exchange integral has no classical analogy. The

electronic energy can thus be written from Eqs. (2.38), (2.39), and (2.40) as

$$\begin{aligned}
E_{el} &= \sum_{i=1}^{N} h_i^{'} + \sum_{i=1}^{N}\sum_{j>i}^{N}(J_{ij}^{'} - K_{ij}^{'}) + V_{nn}, \\
&= \sum_{i=1}^{N}\langle\chi_i(i)|\hat{h}_i|\chi_i(i)\rangle + \sum_{i=1}^{N}\sum_{j>i}^{N}\Big(\langle\chi_i(i)\chi_j(j)|\hat{g}_{ij}|\chi_i(i)\chi_j(j)\rangle \\
&\quad -\langle\chi_i(i)\chi_j(j)|\hat{g}_{ij}|\chi_j(i)\chi_i(j)\rangle\Big) + V_{nn} \\
&= \sum_{i=1}^{N}\langle\chi_i(1)|\hat{h}_1|\chi_i(1)\rangle + \sum_{i=1}^{N}\sum_{j>i}^{N}\Big(\langle\chi_i(1)\chi_j(2)|\hat{g}_{12}|\chi_i(1)\chi_j(2)\rangle \\
&\quad -\langle\chi_i(1)\chi_j(2)|\hat{g}_{12}|\chi_j(1)\chi_i(2)\rangle\Big) + V_{nn} \\
&= \sum_{i=1}^{N} h_i + \sum_{i=1}^{N}\sum_{j>i}^{N}(J_{ij} - K_{ij}) + V_{nn}, \\
&= \sum_{i=1}^{N} h_i + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}(J_{ij} - K_{ij}) + V_{nn}, \qquad (2.43)
\end{aligned}$$

where

$$J_{ij} = \langle\chi_i(1)\chi_j(2)|\hat{g}_{12}|\chi_i(1)\chi_j(2)\rangle, \qquad (2.44)$$

$$K_{ij} = \langle\chi_i(1)\chi_j(2)|\hat{g}_{12}|\chi_j(1)\chi_i(2)\rangle. \qquad (2.45)$$

2.3.4 Lagrange Multipliers

A constrain is just a relationship between the variables and has the form $g(x_1, x_2, \cdots, x_t) =$ constant. To find the extrema of a function $f(x, y)$ subject to a constraint $g(x, y) =$ constant, we solve the two equations simultaneously:

$$df = \Big(\frac{\partial f}{\partial x}\Big)_y dx + \Big(\frac{\partial f}{\partial y}\Big)_x dy = 0, \qquad (2.46a)$$

$$dg = \Big(\frac{\partial g}{\partial x}\Big)_y dx + \Big(\frac{\partial g}{\partial y}\Big)_x dy = 0. \qquad (2.46b)$$

From Eq. (2.46a),

$$\frac{dy}{dx} = \frac{-\partial f/\partial x}{\partial f/\partial y} = \frac{-\partial g/\partial x}{\partial g/\partial y} \qquad (2.47)$$

Because Eq. (2.47) requires the equality of two ratios, the derivatives of f and g need only be the same to within an arbitrary constant λ, called the Lagrange multiplier:

$$\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x} \quad \text{and} \frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y}. \tag{2.48}$$

Another expression for the Lagrange multiplier can be found from Eqs. (2.46a) and (2.48) as

$$df - \lambda dg = \Big(\frac{\partial f}{\partial x} - \lambda \frac{\partial g}{\partial x}\Big) dx + \Big(\frac{\partial f}{\partial y} - \lambda \frac{\partial g}{\partial y}\Big) dy = 0. \tag{2.49}$$

EXAMPLE 3 $f(x, y) = x^2 + y^2$. $x^* = y^* = 0$ is the minimum without any constraint. When a constraint $g(x) = x + y = 6$ is applied, the minimum is $x^* = y^* = 3$.

2.3.5 Minimization of the Energy of a Single Determinant

For the purpose of deriving the variation of the energy, it is convenient to express the energy in Eq. (2.43) in terms of Coulomb ($\hat{J}$) and exchange ($\hat{K}$) operators.

$$E_{el} = \sum_{i=1}^{N} h_i + \frac{1}{2} \sum_{ij} (\langle \chi_j | \hat{J}_i | \chi_j \rangle - \langle \chi_j | \hat{K}_i | \chi_j \rangle) + V_{nn}, \tag{2.50}$$

where

$$\hat{J}_i | \chi_j(2) \rangle = \langle \chi_i(1) | \hat{g}_{12} | \chi_i(1) \rangle | \chi_j(2) \rangle, \tag{2.51}$$

$$\hat{K}_i | \chi_j(2) \rangle = \langle \chi_i(1) | \hat{g}_{12} | \chi_j(1) \rangle | \chi_i(2) \rangle. \tag{2.52}$$

The objective is now to determine a set of MOs that makes the energy a minimum, or at least stationary with respect to a change in the orbitals. The variations, however, must be carried under the constraint that the MOs remain orthonormal. Using Lagrange multiplier, we minimize the Lagrange function with respect to an orbital variation

$$L \equiv E_{el} - \sum_{ij}^{N} \lambda_{ij} (\langle \chi_i | \chi_j \rangle - \delta_{ij}). \tag{2.53}$$

We set the first variation in L equal to zero.

$$\delta L = \delta E_{el} - \sum_{ij}^{N} \lambda_{ij}(\langle \delta\chi_i|\chi_j\rangle + \langle \chi_i|\delta\chi_j\rangle) = 0. \tag{2.54}$$

The variation of the energy E_{el} is given as

$$\begin{aligned}
\delta E &= \sum_i^N (\langle\chi_i|\hat{h}|\chi_i\rangle + \langle\chi_i|\hat{h}_i|\delta\chi_i\rangle) \\
&\quad + \frac{1}{2}\sum_{ij}^N \Big(\langle\delta\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_i(1)\rangle|\chi_j(2)\rangle + \langle\chi_j(2)|\langle\delta\chi_i(1)|\hat{g}_{12}|\chi_i(1)\rangle|\chi_j(2)\rangle \\
&\quad + \langle\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\delta\chi_i(1)\rangle|\chi_j(2)\rangle + \langle\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_i(1)\rangle|\delta\chi_j(2)\rangle \\
&\quad - \langle\delta\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_j(1)\rangle|\chi_i(2)\rangle - \langle\chi_j(2)|\langle\delta\chi_i(1)|\hat{g}_{12}|\chi_j(1)\rangle|\chi_i(2)\rangle \\
&\quad - \langle\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\delta\chi_j(1)\rangle|\chi_i(2)\rangle - \langle\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_j(1)\rangle|\delta\chi_i(2)\rangle\Big) \qquad (2.55) \\
&= \sum_i^N (\langle\chi_i|\hat{h}|\chi_i\rangle + \langle\chi_i|\hat{h}_i|\delta\chi_i\rangle) \\
&\quad + \frac{1}{2}\sum_{ij}^N \Big(\langle\delta\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_i(1)\rangle|\chi_j(2)\rangle + \langle\delta\chi_i(2)|\langle\chi_j(1)|\hat{g}_{12}|\chi_j(1)\rangle|\chi_i(2)\rangle \\
&\quad + \langle\chi_i(2)|\langle\chi_j(1)|\hat{g}_{12}|\chi_j(1)\rangle|\delta\chi_i(2)\rangle + \langle\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_i(1)\rangle|\delta\chi_j(2)\rangle \\
&\quad - \langle\delta\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_j(1)\rangle|\chi_i(2)\rangle - \langle\delta\chi_i(2)|\langle\chi_j(1)|\hat{g}_{12}|\chi_i(1)\rangle|\chi_j(2)\rangle \\
&\quad - \langle\chi_i(2)|\langle\chi_j(1)|\hat{g}_{12}|\chi_i(1)\rangle|\delta\chi_j(2)\rangle - \langle\chi_j(2)|\langle\chi_i(1)|\hat{g}_{12}|\chi_j(1)\rangle|\delta\chi_i(2)\rangle\Big) \qquad (2.56) \\
&= \sum_i^N (\langle\chi_i|\hat{h}|\chi_i\rangle + \langle\chi_i|\hat{h}_i|\delta\chi_i\rangle) \\
&\quad + \frac{1}{2}\sum_{ij}^N \Big(\langle\delta\chi_j(2)|\hat{J}_i|\chi_j(2)\rangle + \langle\delta\chi_i(2)|\hat{J}_j|\chi_i(2) + \langle\chi_i(2)|\hat{J}_j|\delta\chi_i(2)\rangle + \langle\chi_j(2)|\hat{J}_i|\delta\chi_j(2)\rangle \\
&\quad - \langle\delta\chi_j(2)|\hat{K}_i|\chi_i(2)\rangle - \langle\delta\chi_i(2)|\hat{K}_j|\chi_j(2)\rangle - \langle\chi_i(2)|\hat{K}_j|\delta\chi_j(2)\rangle - \langle\chi_j(2)|\hat{K}_i|\delta\chi_i(2)\rangle\Big) \qquad (2.57) \\
&= \sum_i^N (\langle\chi_i|\hat{h}|\chi_i\rangle + \langle\chi_i|\hat{h}_i|\delta\chi_i\rangle) \\
&\quad + \frac{1}{2}\sum_{ij}^N \Big(\langle\delta\chi_i|\hat{J}_j - \hat{K}_j|\chi_i\rangle + \langle\chi_i|\hat{J}_j - \hat{K}_j|\delta\chi_i\rangle + \langle\delta\chi_j|\hat{J}_i - \hat{k}_i|\chi_j\rangle + \langle\chi_j|\hat{J}_i - \hat{K}_i|\delta\chi_j\rangle\Big). \qquad (2.58) \\
&= \sum_i^N (\langle\chi_i|\hat{h}_i|\chi_i\rangle + \langle\chi_i|\hat{h}_i|\delta\chi_i\rangle) + \sum_{ij}^N \Big(\langle\delta\chi_i|\hat{J}_j - \hat{K}_j|\chi_i\rangle + \langle\chi_i|\hat{J}_j - \hat{K}_j|\delta\chi_i\rangle\Big) \qquad (2.59) \\
&= \sum_i^N (\langle\delta\chi_i|\hat{F}_i|\chi_i\rangle + \langle\chi_i|\hat{F}_i|\delta\chi_i\rangle), \qquad (2.60)
\end{aligned}$$

where the Fock operator is introduced as

$$\hat{F}_i = \hat{h}_i + \sum_{j=1}^{N} (\hat{J}_j - \hat{K}_j). \tag{2.61}$$

The Fock operator is an effective one-electron energy operator, describing the kinetic energy of an electron and the attraction to all the nuclei ($\hat{h}_i$), as well as the repulsive to all the other electrons (via the $\hat{J}_i$ and $\hat{K}_i$ operators. Note that the Fock operator is associated with the variation of the total energy, not the energy itself. The electronic Hamiltonian operator $\hat{H}_{el}$ is not a sum of Fock operators. The variation of the Lagrange function L now becomes

$$\delta L = \sum_{i}^{N} (\langle \delta\chi_i | \hat{F}_i | \chi_i \rangle + \langle \chi_i | \hat{F}_i | \delta\chi_i \rangle) - \sum_{ij}^{N} \lambda_{ij} (\langle \delta\chi_i | \chi_j \rangle + \langle \chi_i | \delta\chi_j \rangle). \tag{2.62}$$

The variation principle states that the desired orbitals are those that make $\delta L = 0$. Because

$$\begin{aligned} \langle \chi_i | \delta\chi_j \rangle &= \langle \delta\chi_j | \chi_i \rangle^* \\ \langle \chi_i | \hat{F} | \delta\chi_j \rangle &= \langle \delta\chi_j | \hat{F} | \chi_i \rangle^*, \end{aligned} \tag{2.63}$$

Eq. (2.62) becomes

$$\delta L = \sum_{i}^{N} (\langle \delta\chi_i | \hat{F}_i | \chi_i \rangle - \sum_{ij}^{N} \lambda_{ij} \langle \delta\chi_i | \chi_j \rangle + \langle \delta\chi_i | \hat{F}_i | \chi_i \rangle^* - \sum_{ij}^{N} \lambda_{ij} (\langle \delta\chi_j | \chi_i \rangle^* = 0. \tag{2.64}$$

Since $\langle \delta\chi |$ and $\langle \delta\chi |^*$ arbitrary, $\delta L = 0$ leads to

$$\sum_{i}^{N} \langle \delta\chi_i | (\hat{F}_i | \chi_i \rangle - \sum_{j}^{N} \lambda_{ij} | \chi_j \rangle) = 0, \tag{2.65}$$

$$+ \Big(\langle \delta\chi_i | (\hat{F}_i | \chi_i \rangle - \sum_{j}^{N} \lambda_{ji}^* | \chi_j \rangle) \Big)^* = 0. \tag{2.66}$$

From the equations above,

$$\begin{aligned} \hat{F}_i|\chi_i\rangle &= \sum_j^N \lambda_{ij}|\chi_j\rangle = \lambda_{ji}^*|\chi_j\rangle, \\ &\Rightarrow \sum_j^N (\lambda_{ij} - \lambda_{ji}^*)|\chi_j\rangle, \end{aligned} \tag{2.67}$$

This means that the Lagrange multiplier are elements of a Hermitian matrix ($\lambda_{ij} = \lambda_{ji}^*$). The final set of Hartree-Fock equations may be written as

$$\hat{F}_i|\chi_i\rangle = \sum_j^N \lambda_{ij}|\chi_j\rangle. \tag{2.68}$$

Eq. (2.64) can be simplified by choosing a unitary transformation that makes the matrix of Lagrange multipliers diagonal, (i.e., $\lambda_{ij} = 0$ and $\lambda_{ii} = \epsilon_i$). This special set of molecular orbitals ($\chi^{'}$) is called **canonical** MOs and transforms Eq. (2.68) into

$$\boxed{\hat{F}_i|\chi_i^{'}\rangle = \epsilon_i|\chi_j^{'}\rangle.} \tag{2.69}$$

From Eqs. (2.61) and (2.69),

$$\epsilon_i = \langle\chi_i|\hat{F}_i|\chi_i\rangle = h_i + \sum_j^N (J_{ij} - K_{ij}), \tag{2.70}$$

and the electronic energy is given from Eq. (2.50) as

$$\begin{aligned} E_{el} &= \sum_{i=1}^N h_i + \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij}) + V_{nn}, \\ &= \sum_{i=1}^N \epsilon_i - \frac{1}{2}\sum_{ij}(J_{ij} - K_{ij}) + V_{nn}. \end{aligned} \tag{2.71}$$

The electronic energy is not simply a sum of MO orbital energies. The Fock operator contains terms describing the repulsion to all other electrons, and the sum over MO energies therefore counts the electron-electron repulsion twice. The electron-electron

repulsion is only accounted for in an average fashion, and the HF method is therefore also referred to as a mean field approximation. This is due to the approximation of a single Slater determinant as the trial wave function.

2.3.6 Koopmans' Theorem

Consider the energy of an N-electron system and the corresponding system with one electron removed from orbital number k, and assume that the MOs are identical for the two systems. Then the electronic energies for the two systems are respectively

$$E_N = \sum_{i=1}^{N} h_i + \frac{1}{2}\sum_{i}^{N}\sum_{j}^{N}(J_{ij} - K_{ij}) + V_{nn}, \tag{2.72}$$

$$E_{N-1} = \sum_{i=1}^{N-1} h_i + \frac{1}{2}\sum_{i}^{N-1}\sum_{j=1}^{N-1}(J_{ij} - K_{ij}) + V_{nn}. \tag{2.73}$$

Substracting the two electronic energy gives

$$\begin{aligned} E_N - E_{N-1} &= h_k + \frac{1}{2}\sum_{i=1}^{N}(J_{ik} - K_{ik}) + \frac{1}{2}\sum_{j=1}^{N}(J_{kj} - K_{kj}), \\ &= h_k + \sum_{i=1}^{N}(J_{ik} - K_{ik}) = \epsilon_k. \end{aligned} \tag{2.74}$$

This is exactly the orbital energy ϵ_k. The ionization energy within the "frozen MO" approximation is given simply as the orbital energy, a result known as **Koopmans' theorem**. Similarly, the electron affinity of a neutral molecule is given as the orbital energy of the corresponding anion:

$$E_{N+1}^{k} - E_N = \epsilon_k \tag{2.75}$$

2.4 Restricted Closed-Shell Hartree-Fock: The Roothan Equations

2.4.1 Restricted and Unrestricted Hartree Fock

1. Closed-shell wave function (or determinant): A determinant in which each spatial orbital is doubly occupied. A singlet type of wave function.

2. Open-shell wave function (or determinant): A determinant in which at least one spatial orbital is singly occupied. A doublet or triplet type of wave function.

3. Restricted Hartree-Fock wave function: If there is the restriction that each spatial orbital should have two electrons, one with α and one with β spin, the trial function is a Restricted Hartree-Fock (RHF) wave function.

   Given a set of K orthonormal spatial orbitals $\{\psi_i | i = 1, 2, \cdots, K\}$, we can form a set of $2K$ spin orbitals $\{\chi_i | i = 1, 2, \cdots, 2K\}$ by multiplying each spatial orbital by either the α or β spin function

$$\begin{aligned} \chi_{2i-1} &= \psi_i(\mathrm{r})\alpha(\omega) \\ \chi_{2i} &= \psi_i(\mathrm{r})\beta(\omega) \quad i = 1, 2, \cdots, K. \end{aligned} \tag{2.76}$$

4. Unrestricted Hartree-Fock (UHF) wave function: If there are no restrictions on the form of the spatial orbitals, the trial function is an Unrestricted Hartree-Fock (UHF) wave function.

   Given a set of K orthonormal spatial orbitals $\{\psi_i^\alpha | i = 1, 2, \cdots, K\}$,

$$\langle \psi_i^\alpha | \psi_j^\alpha \rangle = \delta_{ij}, \tag{2.77}$$

   and a different set of K orthonormal spatial orbitals $\{\psi_i^\beta | i = 1, 2, \cdots, K\}$,

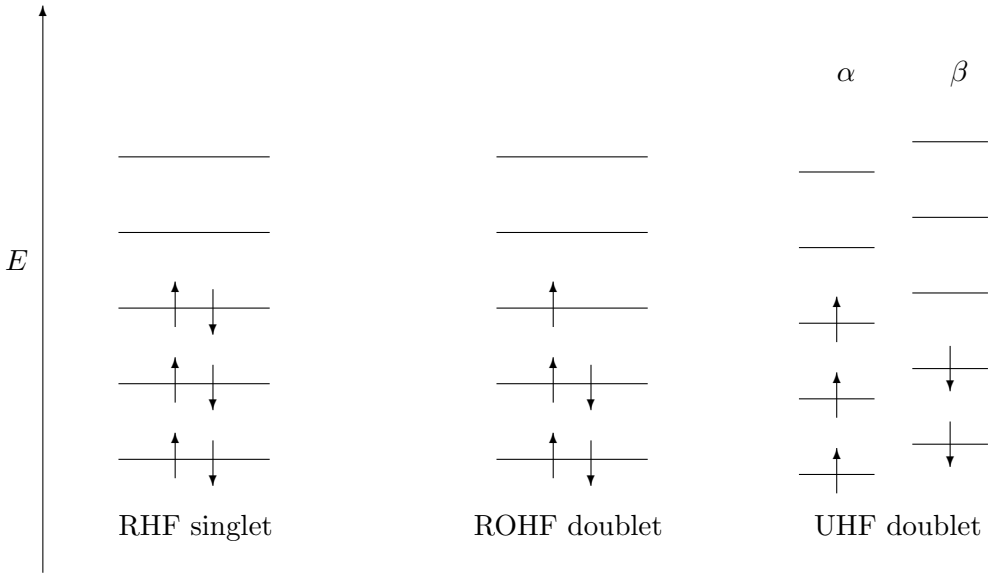
$$\langle \psi_i^\beta | \psi_j^\beta \rangle = \delta_{ij}, \tag{2.78}$$

such that the two sets are not orthogonal,

$$\langle \psi_i^\alpha | \psi_j^\beta \rangle = S_{ij}^{\alpha\beta}, \tag{2.79}$$

where $\mathbf{S}^{\alpha\beta}$ is an overlap matrix, we can form $2K$ unrestricted spin orbitals as

$$\begin{aligned} \chi_{2i-1} &= \psi_i^\alpha(\mathrm{r})\alpha(\omega) \\ \chi_{2i} &= \psi_i^\beta(\mathrm{r})\beta(\omega) \quad i = 1, 2, \cdots, K. \end{aligned} \tag{2.80}$$



5. At large bond lengths, the energy is lowered by using an unrestricted rather than a restricted description.

2.4.2 Closed-Shell Hartree-Fock: From Restricted Spin Orbitals to Spatial Orbitals

The wave function (or Slater determinant) for the closed-shell restricted ground state is

$$\Psi_0\rangle = |\chi_1\chi_2\cdots\chi_{N-1}\chi_N\rangle = |\psi_1\bar{\psi}_1\cdots\psi_a\bar{\psi}_a\cdots\psi_{N/2}\bar{\psi}_{N/2}\rangle. \tag{2.81}$$

We now want to convert the general spin orbital Hartree-Fock equation $\hat{F}(\mathbf{x}_1)\chi_i(\mathbf{x}_1) = \epsilon_i\chi_i(\mathbf{x}_1)$ to a spatial eigenvalue equation where each of the occupied spatial molecular orbitals $\{\psi_a|a = 1, 2, \cdots, N/2\}$ is doubly occupied. The Hartree-Fock equation for a spin orbital is

$$\hat{F}(\mathbf{x}_1)\chi_i(\mathbf{x}_1) = \epsilon_i\chi_i(\mathbf{x}_1). \tag{2.82}$$

The spin orbital $\chi_i(\mathbf{x}_1)$ will have either the α or β spin function. Let us assume α. Then Eq. (2.83) becomes

$$\hat{F}(\mathbf{x}_1)\psi_j(\mathbf{r}_1)\alpha(\omega_1) = \epsilon_i\psi_j(\mathbf{r}_1)\alpha(\omega_1). \tag{2.83}$$

Multiplying on the left by $\alpha^*(\omega_1)$ and integrating over spin gives

$$\left[\int \mathrm{d}\omega_1\alpha^*(\omega_1)\hat{F}(\mathbf{x}_1)\alpha(\omega_1)\right]\psi_j(\mathbf{r}_1) = \epsilon_j\psi_j(\mathbf{r}_1). \tag{2.84}$$

Because

$$\hat{F}(\mathbf{x}_1) = \hat{h}(\mathbf{r}_1) + \sum_c^N \int \mathrm{d}\mathbf{x}_2\chi_c^*(\mathbf{x}_2)r_{12}^{-1}(1 - \hat{P}_{12})\chi_c(\mathbf{x}_2). \tag{2.85}$$

Eq. (2.84) becomes

$$\left[\int \mathrm{d}\omega_1\alpha^*(\omega_1)\hat{F}(\mathbf{x}_1)\alpha(\omega_1)\right]\psi_j(\mathbf{r}_1) = \left[\int \mathrm{d}\omega_1\alpha^*(\omega_1)\hat{h}(\mathbf{r}_1)\alpha(\omega_1)\right]\psi_j(\mathbf{r}_1)$$
$$+\sum_c^N \int \mathrm{d}\omega_1\mathrm{d}\mathbf{x}_2\alpha^*(\omega_1)\chi_c^*(\mathbf{x}_2)r_{12}^{-1}(1 - \hat{P}_{12})\chi_c(\mathbf{x}_2)\alpha(\omega_1)\Big]\psi_j(\mathbf{r}_1) = \epsilon_j\psi_j(\mathbf{r}_1). \tag{2.86}$$

Defining $\hat{F}(\mathbf{r}_1)$ as

$$\hat{F}(\mathbf{r}_1) = \int \mathrm{d}\omega_1\alpha^*(\omega_1)\hat{F}(\mathbf{x}_1)\alpha(\omega_1). \tag{2.87}$$

leads to

$$\begin{aligned}\hat{F}(\mathbf{r}_1)(\mathbf{r}_1) &= \hat{h}(\mathbf{r}_1)\psi_j(\mathbf{r}_1) + \sum_c^N \int \mathrm{d}\omega_1\mathrm{d}\mathbf{x}_2\alpha^*(\omega_1)\chi_c^*(\mathbf{x}_2)r_{12}^{-1}\chi_c(\mathbf{x}_2)\alpha(\omega_1)\psi_j(\mathbf{r}_1) \\ &\quad -\sum_c^N \int \mathrm{d}\omega_1\mathrm{d}\mathbf{x}_2\alpha^*(\omega_1)\chi_c^*(\mathbf{x}_2)r_{12}^{-1}\chi_c(\mathbf{x}_1)\alpha(\omega_2)\psi_j(\mathbf{r}_2) \\ &= \epsilon_j\psi_j(\mathbf{r}_1).\end{aligned} \tag{2.88}$$

If we have a closed-shell, the sum over occupied spin orbitals includes an equal sum over those with α spin function and those with β spin function and Eq. (2.88) can be written as

$$\begin{aligned}
\hat{F}(\mathbf{r}_1)\psi_j(\mathbf{r}_1) &= \hat{h}(\mathbf{r}_1)\psi_j(\mathbf{r}_1) \\
&\quad + \sum_c^{N/2} \int \mathrm{d}\omega_1 \mathrm{d}\omega_2 \mathrm{d}\mathbf{r}_2 \alpha^*(\omega_1)\psi_c^*(\mathbf{r}_2)\alpha(\omega_2) r_{12}^{-1} \psi_c(\mathbf{r}_2)\alpha(\omega_2)\alpha(\omega_1)\psi_j(\mathbf{r}_1) \\
&\quad + \sum_c^{N/2} \int \mathrm{d}\omega_1 \mathrm{d}\omega_2 \mathrm{d}\mathbf{r}_2 \alpha^*(\omega_1)\psi_c^*(\mathbf{r}_2)\beta(\omega_2) r_{12}^{-1} \psi_c(\mathbf{r}_2)\beta(\omega_2)\alpha(\omega_1)\psi_j(\mathbf{r}_1) \\
&\quad - \sum_c^{N/2} \int \mathrm{d}\omega_1 \mathrm{d}\omega_2 \mathrm{d}\mathbf{r}_2 \alpha^*(\omega_1)\psi_c^*(\mathbf{r}_2)\alpha(\omega_2) r_{12}^{-1} \psi_c(\mathbf{r}_1)\alpha(\omega_1)\alpha(\omega_2)\psi_j(\mathbf{r}_2) \\
&\quad - \sum_c^{N/2} \int \mathrm{d}\omega_1 \mathrm{d}\omega_2 \mathrm{d}\mathbf{r}_2 \alpha^*(\omega_1)\psi_c^*(\mathbf{r}_2)\beta(\omega_2) r_{12}^{-1} \psi_c(\mathbf{r}_1)\beta(\omega_1)\alpha(\omega_2)\psi_j(\mathbf{r}_2) \\
&= \hat{h}(\mathbf{r}_1)\psi_j(\mathbf{r}_1) + \left[2 \sum_c^{N/2} \int \mathrm{d}\mathbf{r}_2 \psi_c^*(\mathbf{r}_2) r_{12}^{-1} \psi_c(\mathbf{r}_2) \right] \psi_j(\mathbf{r}_1) \\
&\quad - \left[\sum_c^{N/2} \int \mathrm{d}\mathbf{r}_2 \psi_c^*(\mathbf{r}_2) r_{12}^{-1} \psi_j(\mathbf{r}_2) \right] \psi_c(\mathbf{r}_1) \\
&= \epsilon_j \psi_j(\mathbf{r}_1). \qquad (2.89)
\end{aligned}$$

The closed-shell Fock operator thus has the form,

$$\hat{F}(\mathbf{r}_1) = \hat{h}(\mathbf{r}_1) + \sum_a^{N/2} \int \mathrm{d}\mathbf{r}_2 \psi_a^*(\mathbf{r}_2) r_{12}^{-1} (2 - \hat{P}_{12}) \psi_a(\mathbf{r}_2), \qquad (2.90)$$

$$\hat{F}(1) = \hat{h}(1) + \sum_a^{N/2} 2\hat{J}_a(1) - \hat{K}_a(1), \qquad (2.91)$$

where the closed-shell Coulomb and exchange operators are defined by

$$\hat{J}_a(1) = \int \mathrm{d}\mathbf{r}_2 \psi_a^*(2) r_{12}^{-1} \psi_a(2) \qquad (2.92)$$

$$\hat{K}_a(1)\psi_i(1) = \left[\int \mathrm{d}\mathbf{r}_2 \psi_a^*(2) r_{12}^{-1} \psi_i(2) \right] \psi_a(1). \qquad (2.93)$$

The closed-shell spatial Hartree-Fock equation is just

$$\hat{F}(1)\psi_j(1) = \epsilon_j\psi_j(1). \tag{2.94}$$

The closed-shell orbital energy is then given as

$$\begin{aligned} \epsilon_i &= h_i + \sum_j^N (J_{ij} - K_{ij}), & (2.95)\\ &= h_i + \sum_b^{N/2} (2J_{ib} - K_{ib}), & (2.96) \end{aligned}$$

and the closed-shell Hartree-Fock energy is then given as

$$\begin{aligned} E_{el} &= \sum_{i=1}^N h_i + \frac{1}{2}\sum_{i=1}^N\sum_{j=1}^N (J_{ij} - K_{ij}) + V_{nn}, & (2.97)\\ &= 2\sum_a^{N/2} h_a + \sum_a^{N/2}\sum_b^{N/2} (2J_{ab} - K_{ab}), & (2.98)\\ &= 2\sum_a^{N/2} \epsilon_a - \sum_a^{N/2}\sum_b^{N/2} (2J_{ab} - K_{ab}). & (2.99) \end{aligned}$$

2.4.3 Introduction of a Basis: The Roothan-(Hall) Equations

To solve the closed-shell spatial Hartree-Fock equation

$$\hat{F}(\mathbf{r}_1)\psi_j(\mathbf{r}_1) = \epsilon_j\psi_j(\mathbf{r}_1), \tag{2.100}$$

we introduce a set of K known basis functions $\{\phi_\mu(\mathbf{r})|\mu = 1, 2, \cdots, K\}$, conventionally called **atomic orbitals**. The basis functions are assumed to be normalized and linearly independent, but are not in general orthogonal to each other. The linear expansion of each MO in terms of the basis functions (LCAO-MO, Linear Combination of Atomic Orbitals Molecular Orbital) leads to

$$\psi_j(\mathbf{r}_1) = \sum_{\mu=1}^K C_{\mu j}\phi_\mu(\mathbf{r}_1), \quad j = 1, 2, \cdots, K. \tag{2.101}$$

Then, the Hartree-Fock equations Eq. (2.100) may be written as

$$\hat{F}(1)\sum_{\nu=1}^{K} C_{\nu j}\phi_\nu(1) = \epsilon_j \sum_{\nu=1}^{K} C_{\nu i}\phi_\nu(1). \tag{2.102}$$

By multiplying by $\phi_\mu^*(1)$ on the left and integrating, we turn the integro-differential equation into a matrix equation,

$$\sum_{\nu=1}^{K} C_{\nu j}\int d\mathbf{r}_1 \phi_\mu^* \hat{F}(1)\phi_\nu(1) = \epsilon_j \sum_{\nu=1}^{K} C_{\nu i}\int d\mathbf{r}_1 \phi_\mu^*(1)\phi_\nu(1) \tag{2.103}$$

Now we define two matrices.

1. The **overlap matrix S** has elements

$$S_{\mu\nu} = \int d\mathbf{r}_1 \phi_\mu^*(1)\phi_\nu(1), \tag{2.104}$$

   and is a $K \times K$ Hermitian (although usually real and symmetric) matrix. Since the basis functions are normalized, but are not orthogonal to each other, the diagonal elements of **S** unity and the off-diagonal elements are numbers less than one in magnitude, i.e., $0 \leq |S_{\mu\nu}| \leq 1$.

2. The **Fock matrix F** has elements

$$F_{\mu\nu} = \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{F}(1)\phi_\nu(1), \tag{2.105}$$

   and is also a $K \times K$ Hermitian (although usually real and symmetric) matrix.

With these definitions of **F** and **S**, we can now write Eq. (2.103) as

$$\sum_\nu F_{\mu\nu}C_{\nu j} = \epsilon_j \sum_\nu S_{\mu\nu}C_{\nu j}, \quad i = 1, 2, \cdots, K. \tag{2.106}$$

Eq. (2.106) is called the **Roothaan-(Hall) equations**, which can be written more compactly as the single matrix equation:

$$\mathbf{FC} = \mathbf{SC}\boldsymbol{\epsilon}, \tag{2.107}$$

where $\mathbf{C}$ is a $K \times K$ matrix of the expansion coefficients $C_{\mu j}$:

$$\mathbf{C} = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1K} \\ C_{21} & C_{22} & \cdots & C_{2K} \\ \vdots & \vdots & & \vdots \\ C_{K1} & C_{K2} & \cdots & C_{KK} \end{pmatrix}, \tag{2.108}$$

(2.109)

and $\boldsymbol{\epsilon}$ is a diagonal matrix of the orbital energies ϵ_j:

$$\boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 & 0 & \cdots & 0 \\ 0 & \epsilon_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \epsilon_K \end{pmatrix}. \tag{2.110}$$

(2.111)

The Fock matrix $\mathbf{F}$ is the matrix representation of the Fock operator and for the closed-shell system, is written as

$$\hat{F}(\mathbf{r}_1) = \hat{h}(\mathbf{r}_1) + \sum_a^{N/2} \int \mathrm{d}\mathbf{r}_2 \psi_a^*(\mathbf{r}_2) r_{12}^{-1} (2 - \hat{P}_{12}) \psi_a(\mathbf{r}_2), \tag{2.112}$$

or

$$\hat{F}(1) = \hat{h}(1) + \sum_a^{N/2} 2\hat{J}_a(1) - \hat{K}_a(1). \tag{2.113}$$

From Eq. (2.113), the elements of the Fock matrix $F_{\mu\nu}$ can be written as

$$
\begin{aligned}
F_{\mu\nu} &= \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{F}(1)\phi_\nu(1), && (2.114)\\
&= \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{h}(1)\phi_\nu(1) + \sum_a^{N/2} \int d\mathbf{r}_1 \phi_\mu^*(1)[2\hat{J}_a(1) - \hat{K}_a(1)]\phi_\nu(1), && (2.115)\\
&= \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{h}(1)\phi_\nu(1) + \sum_a^{N/2} \Bigg(2\int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\psi_a^*(2) r_{12}^{-1} \psi_a(2)\phi_\nu(1)\\
&\quad - \int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\psi_a^*(2) r_{12}^{-1} \hat{P}_{12}\psi_a(2)\phi_\nu(1)\Bigg), && (2.116)\\
&= \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{h}(1)\phi_\nu(1) + \sum_a^{N/2} \Bigg(2\int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\psi_a^*(2) r_{12}^{-1} \psi_a(2)\phi_\nu(1)\\
&\quad - \int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\psi_a^*(2) r_{12}^{-1} \phi_\nu(2)\psi_a(1)\Bigg), && (2.117)\\
&= \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{h}(1)\phi_\nu(1) + \sum_a^{N/2} \Bigg(2\int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\psi_a^*(2) r_{12}^{-1} \psi_a(2)\phi_\nu(1)\\
&\quad - \int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\psi_a^*(2) r_{12}^{-1} \phi_\nu(2)\psi_a(1)\Bigg), && (2.118)\\
&= \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{h}(1)\phi_\nu(1) + \sum_a^{N/2}\sum_{\lambda\sigma} C_{\sigma a}^* C_{\lambda a} \Bigg(2\int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\phi_\sigma^*(2) r_{12}^{-1} \phi_\lambda(2)\phi_\nu(1)\\
&\quad - \int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\phi_\sigma^*(2) r_{12}^{-1} \phi_\nu(2)\phi_\lambda(1)\Bigg), \quad \Big(\psi_a(2) = \sum_{\lambda=1}^{K} C_{\lambda a}\phi_\lambda(2)\Big). && (2.119)
\end{aligned}
$$

There are two notations for two-electron integrals. One is the physicist's notation, where the ordering of the functions is given as

$$\langle \mu\sigma|\nu\lambda\rangle = \int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\phi_\sigma^*(2) r_{12}^{-1}\phi_\nu(1)\phi_\lambda(2). \tag{2.120}$$

The other one is the Mulliken or chemist's notation, which is given as

$$(\mu\nu|\sigma\lambda) = \int d\mathbf{r}_1\mathbf{r}_2 \phi_\mu^*(1)\phi_\nu^*(1) r_{12}^{-1}\phi_\sigma(2)\phi_\lambda(2). \tag{2.121}$$

With the chemist's notation, Eq. (2.119) can be written as

$$F_{\mu\nu} = \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{h}(1)\phi_\nu(1) + \sum_a^{N/2}\sum_{\lambda\sigma} C_{\lambda a}C_{\sigma a}^*\Big[2(\mu\nu|\sigma\lambda) - (\mu\lambda|\sigma\nu)\Big], \quad (2.122)$$

$$= H_{\mu\nu}^{\text{core}} + \sum_{\lambda\sigma} P_{\lambda\sigma}\Big[(\mu\nu|\sigma\lambda) - \frac{1}{2}(\mu\lambda|\sigma\nu)\Big]. \quad (2.123)$$

Here we have defined a core-Hamiltonian matrix as

$$H_{\mu\nu}^{\text{core}} = \int d\mathbf{r}_1 \phi_\mu^*(1)\hat{h}(1)\phi_\nu(1), \quad (2.124)$$

where

$$h(1) = -\frac{1}{2}\nabla_1^2 - \sum_{A=1}^{M}\frac{Z_A}{|\mathbf{r}_1 - \mathbf{R}_A|}. \quad (2.125)$$

We have also defined a **density matrix** or charge-density bond-order matrix as

$$P_{\lambda\sigma} = 2\sum_{a=1}^{N/2} C_{\lambda a}C_{\sigma a}^*. \quad (2.126)$$

Eq. (2.123) can be simplified as

$$F_{\mu\nu} = H_{\mu\nu}^{\text{core}} + G_{\mu\nu}, \quad (2.127)$$

where $G_{\mu\nu}$ is the two-electron part of the Fock matrix. Because the Fock matrix depends on the density matrix,

$$\mathbf{F} = \mathbf{F}(\mathbf{P}) = \mathbf{F}(\mathbf{C}), \quad (2.128)$$

and the Roothan-Hall equations are nonlinear, i.e.,

$$\mathbf{F}(\mathbf{C})\mathbf{C} = \mathbf{S}\mathbf{C}\boldsymbol{\epsilon}. \quad (2.129)$$

If $\mathbf{S}$ were the unit matrix (i.e., if the basis functions are orthonormal), then we would have

$$\mathbf{F}(\mathbf{C})\mathbf{C} = \mathbf{C}\boldsymbol{\epsilon}, \quad (2.130)$$

and the Roothan-Hall equations would just have the form of the usual matrix eigenvalue problem.

2.5 Self-Consistent Field Techniques

2.5.1 SCF procedure

1. Calculate all one- and two-electron integrals and save them in the disk space or memory.

2. Generate a suitable start guess for the MO coefficients → initial guess orbitals.

3. Form the initial density matrix.

4. Form the Fock matrix as the core (one-electron) integrals + the density matrix times the two-electron integrals.

5. Diagonalize the Fock matrix. The eigenvectors contain the new MO coefficients.

6. Form the new density matrix. If it is sufficiently close to the previous density matrix, we are done, otherwise go to step 4.